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POS-Native MLOps: An Accessible Five-Layer Campaign Intelligence Framework for Small and Medium Enterprises in Retail and Food Service

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Abstract

Small and medium enterprises (SMEs) in retail and food service routinely collect transactional data through point-of-sale (POS) systems yet rarely leverage it for marketing decisions, primarily due to resource constraints and the technical complexity of existing analytics solutions. This paper presents a five-layer MLOps pipeline — the **ML Campaign Engine** — that transforms raw POS transaction records requiring only six standard fields into financially validated, per-customer campaign action plans without requiring a dedicated data science team. The framework integrates RFM-based customer segmentation (K-means with dynamic cluster selection via silhouette scoring), market basket analysis (Apriori), dynamic context detection, uplift modeling (LightGBM), and a local large language model (LLM) reporting layer. A novel **CFO Guardrail** mechanism

enforces a configurable minimum net margin floor prior to campaign dispatch, ensuring that no promotion is deployed at a financial loss regardless of predicted uplift score — an architectural contribution that embeds financial constraint validation directly into the ML decision loop. The framework is evaluated on two datasets: (1) a synthetic restaurant POS dataset calibrated to F&B SME industry benchmarks (9,344 transaction line items, 200 customers, 6-month observation window; silhouette = 0.679; k = 2); and (2) a public real-world marketing campaign dataset containing ground-truth promotion response labels (905 customers, aggregated spending records; silhouette = 0.347; k = 3). A T-learner uplift model achieves an AUUC of 0.2351 on the synthetic dataset, outperforming random targeting at identifying high-responders. The CFO Guardrail prevents an estimated ฿2,695 in negative expected profit relative to an unguarded pipeline, demonstrating its role as the primary financial safety mechanism. Ablation analysis confirms that the CFO Guardrail is the primary mechanism preventing money-losing campaigns on this synthetic dataset: removing it reduces total expected profit by ฿2,695; removing the uplift model reduces it by ฿2,162; removing context detection has no material impact on the synthetic dataset. The full pipeline outperforms naive broadcast targeting by +0.3% in net incremental profit on the 200-customer synthetic dataset — a modest differential reflecting the dataset's small scale; the larger improvement (+฿3,496) vs. RFM-only targeting without an uplift model demonstrates the pipeline's value at this scale. Results demonstrate that the proposed framework is robust across F&B and retail contexts, computationally accessible to SME hardware, and architecturally extensible to additional business verticals.

Keywords: MLOps, SME marketing, uplift modeling, RFM segmentation, market basket analysis, campaign optimization, POS analytics, LightGBM, CFO guardrail, Champion-Challenger

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1. Introduction

Marketing personalization has become a competitive necessity in modern retail. Large enterprises routinely deploy dedicated analytics teams, expensive CRM platforms, and data warehouses to generate campaign recommendations at the individual customer level. For small and medium enterprises (SMEs) — which constitute the majority of businesses in most economies — this infrastructure remains economically and

technically out of reach. The consequence is that SME operators make campaign decisions based on intuition or blanket promotions, systematically under-leveraging the transactional data they already possess.

This resource asymmetry creates a measurable market opportunity. POS systems are near-ubiquitous in modern retail and food service, and the transaction data they generate contains rich signals about customer value, product affinity, and purchasing patterns. If an automated pipeline could consume the POS export that any modern cash register already produces and return campaign actions validated against the operator's own financial constraints, the marginal cost of personalized marketing could approach zero for SMEs.

The system described in this paper, referred to as the **ML Campaign Engine**, was designed to address this gap. Its design is governed by four principles:

1. **Minimal input barrier** — the only required data is a standard six-column POS export (`transaction_id`, `customer_id`, `timestamp`, `total_price`, `quantity`, `product_name`). No additional data infrastructure, tagging, or transformation is required from the operator.
2. **Financial safety by default** — every campaign recommendation is blocked if it violates a configurable net margin floor, decoupling ML optimization from financial risk without requiring a human review step.
3. **Operator-readable output** — a local LLM layer translates model outputs into plain-language executive summaries suitable for business owners without a data background.
4. **Production-grade robustness** — a Champion-Challenger update protocol and a six-scenario stress test suite ensure reliable behavior under distribution shifts common in SME operations.

1.1 Research Objectives

The primary objective of this work is:

> *To design and validate a minimal-input MLOps pipeline that enables SMEs in retail and food service to generate per-customer campaign action plans with integrated financial constraint validation, without requiring dedicated data science expertise.*

Specific objectives are:

1. **To design** a five-layer pipeline architecture that accepts standard six-column POS exports without additional infrastructure requirements.
2. **To develop** a CFO Guardrail mechanism that integrates net margin validation directly into the ML campaign decision loop.
3. **To evaluate** segmentation quality and campaign profitability on a synthetic F&B dataset and a public real-world marketing campaign dataset.
4. **To describe and implement** the Champion-Challenger model update protocol and assess its behavior on synthetic data, and ablate each pipeline component to quantify its individual contribution.
5. **To demonstrate** the extensibility of the framework across business verticals through configurable business parameters.

1.2 Contributions

The primary contributions of this work are:

- A **five-layer MLOps architecture** designed from the ground up for SME operational constraints (Section 3).
- A **CFO Guardrail** component that integrates financial validation directly into the ML decision loop, eliminating a manual organizational review step (Section 4.4).
- **Empirical evaluation** on a synthetic F&B dataset (200 customers, silhouette = 0.679, AUUC = 0.2351) and a public real-world marketing campaign dataset (905 customers), including ablation study, baseline comparison, and multi-seed stability analysis (Section 6).
- A **Champion-Challenger protocol** adapted for SME environments with limited historical data (Section 4.4).

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2. Related Work

2.1 RFM Segmentation

Recency-Frequency-Monetary (RFM) analysis is a well-established customer segmentation technique in direct marketing, originally formalized by Hughes [1] and subsequently refined for database marketing applications by Bult and Wansbeek [2].

Classical RFM assigns customers to decile ranks across each dimension; more recent work by Chen et al. [3] augments this with clustering algorithms to capture non-linear segment boundaries that decile ranking cannot express. Our Layer 1 extends classical RFM with K-means clustering [4] and dynamic cluster selection via silhouette scoring [5], and adds an IQR-based outlier filter [6] to prevent high-spend anomalies from distorting segment centroids — a practical concern in SME datasets where a single bulk transaction can represent a substantial fraction of total revenue.

2.2 Market Basket Analysis

Association rule mining, particularly the Apriori algorithm introduced by Agrawal and Srikant [7], has been widely applied in retail to identify product affinity. The algorithmic framework was later made accessible to practitioners through software implementations such as arules [8]. Prior work focuses largely on large-scale grocery and e-commerce datasets [9]. Application to SME food service and mixed retail — where transaction volumes are lower and basket sizes smaller — remains underexplored. Our Layer 2 addresses this by operating at a minimum support threshold of 0.01, appropriate for smaller datasets, and falls back gracefully when no product category column is present in the POS export.

2.3 Uplift Modeling

Uplift modeling (also called incremental response modeling or true-lift modeling) estimates the causal effect of a campaign on an individual customer, rather than the probability of conversion in the presence of a campaign — a distinction first formally articulated by Radcliffe and Surry [10]. A comprehensive review by Devriendt et al. [12] surveys the state of the art across tree-based, regression, and meta-learner approaches. Gutierrez and Gérardy [11] further connect uplift modeling to causal inference frameworks. Gradient boosted tree models, including XGBoost [15] and LightGBM [14], have shown strong performance in uplift tasks [12]; decision-tree-based uplift methods were systematically compared by Rzepakowski and Jaroszewicz [13]. Existing literature focuses predominantly on e-commerce and subscription contexts. Application to POS-driven SME campaigns — with the additional constraint of per-customer margin validation — is a novel contribution of this work.

2.4 MLOps for SMEs

The MLOps literature has grown substantially around large-scale deployments, with Sculley et al. [16] identifying the hidden technical debt incurred when ML systems are embedded in production software, and Kreuzberger et al. [17] providing a formal definition and architecture taxonomy for MLOps. Paleyes et al. [18] survey real-world deployment challenges across a large body of case studies. Work specifically addressing MLOps design for SME constraints — limited compute, small datasets, absence of dedicated ML engineers — remains sparse. Model governance practices including Champion-Challenger selection are discussed in Shmueli et al. [19]. Fairness and bias considerations in ML systems — relevant to equitable campaign targeting across customer segments — are addressed in Barocas et al. [20]. Our Champion-Challenger protocol extends governance practice with a financially motivated update gate that promotes a challenger model only when it demonstrates measurable profit improvement over the incumbent champion.

2.5 LLM Integration in Business Pipelines

Large language model capabilities and limitations are surveyed in Zhao et al. [22], while hallucination risks in generative systems are systematically catalogued by Ji et al. [23] — motivating our structured post-generation validation step. We adopt a local, quantized LLM (llama3.2:3b via Ollama) [21] to ensure that no customer transaction data leaves the operator's infrastructure, addressing data privacy concerns relevant to SME deployments. The local inference approach trades generation quality for zero ongoing inference cost and full data sovereignty — a trade-off appropriate for SME contexts.

2.6 SME Technology Adoption

Barriers to technology adoption among SMEs are well documented. Nguyen et al. [25] confirm that perceived complexity and resource constraints are the primary inhibitors of IT adoption in small businesses, consistent with the access gap in marketing analytics that this work addresses. CRM adoption patterns in SME contexts are examined by Harrigan et al. [24], who find that relationship management tools are most effective when they reduce rather than increase operational complexity — a design principle reflected in the plain-language LLM output layer of this framework.

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3. System Architecture

The ML Campaign Engine is a five-layer sequential pipeline. Each layer consumes the output of the previous layer, applies a specific algorithm, and passes enriched customer records downstream. Figure 1 illustrates the overall data flow.

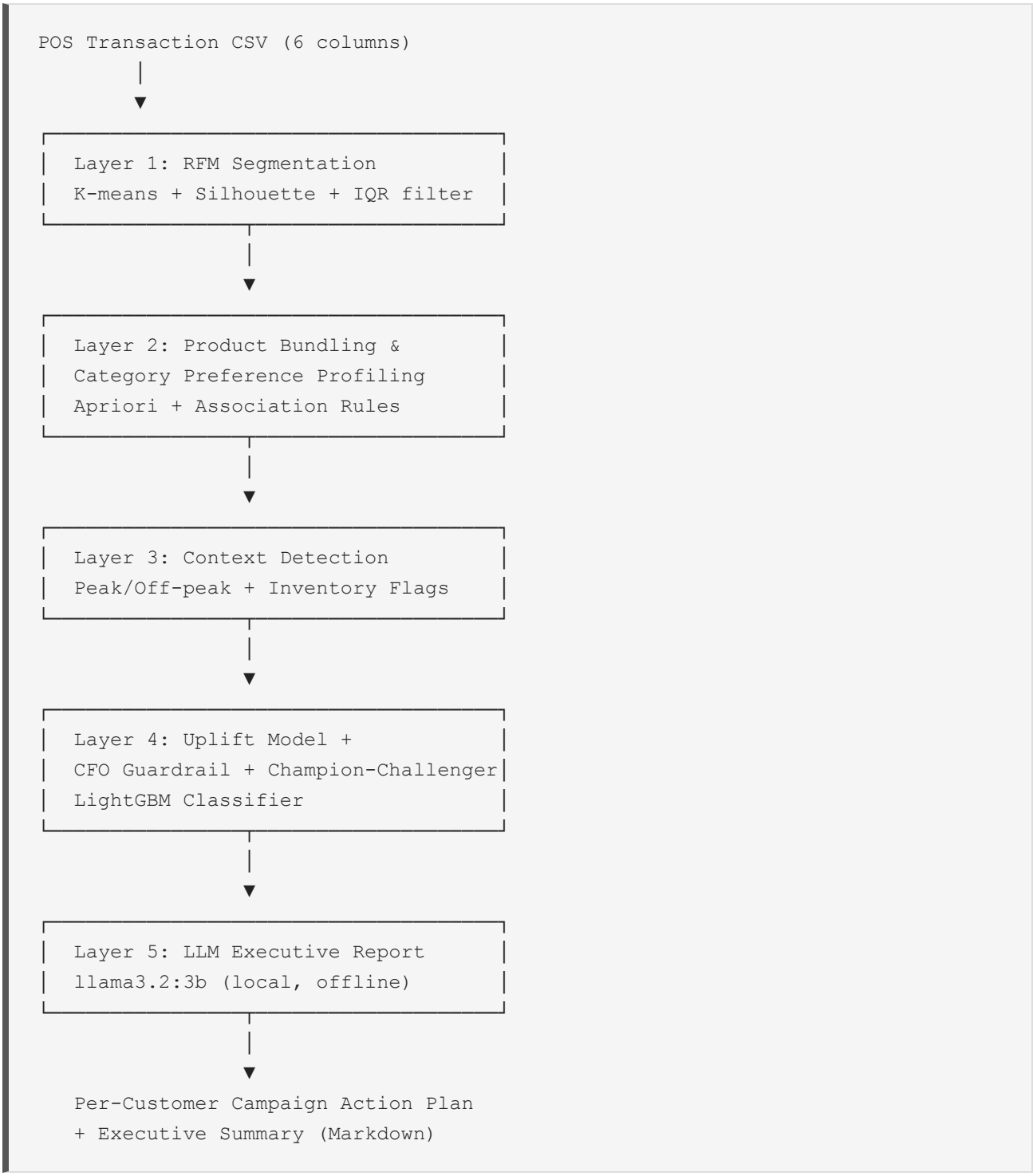


Figure 1. Five-layer ML Campaign Engine pipeline.

The pipeline operates in three modes:

Mode	Entry Point	Use Case
In-memory	`pipeline_engine.py`	Real-time scoring, monthly re-tuning
CSV-driven	`csv_pipeline_engine.py`	Batch processing, audit trail
CFO Guardrail	`run_pipeline.py`	Financial compliance, executive reporting

All business parameters — COGS rate, margin floor, discount rates, peak hours, timezone — are centralized in a single `business_config.py` file, enabling deployment across business types without code changes.

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4. Methodology

4.1 Layer 1 — RFM Customer Segmentation

Input: Raw POS transactions **Output:** Per-customer RFM decile (1–10) and K-means cluster assignment

We compute three metrics per customer from raw transaction history:

- **Recency (R):** Days elapsed since the customer's most recent transaction, relative to the analysis date
- **Frequency (F):** Total count of distinct transactions
- **Monetary (M):** Cumulative spend across all transactions

Outlier filtering: Prior to clustering, we apply Tukey-style outlier filtering on Monetary values. Records with Monetary > Q3 + 3 × IQR are excluded [6], where Q3 is the 75th percentile and IQR = Q3 − Q1. This prevents bulk buyers — common in SME data — from distorting segment centroids.

Normalization: All three dimensions are standardized to zero mean and unit variance (StandardScaler) to prevent Monetary (typically in hundreds) from dominating Recency and Frequency (typically single digits).

Dynamic cluster selection: Cluster count k is selected from $k \in \{2, \ldots, 5\}$ by maximizing the silhouette score:

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

where *a(i)* is the mean intra-cluster distance and *b(i)* is the mean nearest-cluster distance for point *i*. The silhouette score ranges from −1 to +1, with higher values indicating better-defined clusters. Avoiding hardcoded *k* is critical for SME deployment: a boutique café (k=2) and a multi-category retailer (k=5) exhibit structurally different customer distributions.

Segment naming: Clusters are mapped to human-readable tier names by combining monetary rank with recency quadrant: Champions, High Value (Dormant), Potential Loyalists, Need Attention, New/Promising, At-Risk.

4.2 Layer 2 — Market Basket Analysis and Category Profiling

Input: Raw POS transactions **Output:** Top-3 association rules; per-customer category affinity vector

We apply the Apriori algorithm to find frequent product co-purchase patterns. Three metrics govern rule quality:

- **Support:** Fraction of transactions containing both items — set at minimum 0.01 for SME-scale data
- **Confidence:** Conditional probability P(B | A)
- **Lift:** Ratio of observed co-purchase frequency to expected frequency under independence. Lift > 1 indicates positive association; higher lift indicates stronger bundle potential

Rules are ranked by lift and the top three are surfaced for bundle campaign assignment. In parallel, each customer's spend is aggregated by product category to produce a ranked `top_category` and `top_product` affinity profile used by Layer 3.

A graceful fallback is implemented: if the input CSV lacks a `category` column — common in basic POS exports — `product_name` is used as a proxy category, ensuring compatibility with minimally structured data.

4.3 Layer 3 — Context Detection and Campaign Matching

Input: Layer 1 cluster assignments, Layer 2 category profiles, timestamps **Output:** Per-transaction context flags; initial campaign assignment

Temporal context: Timestamps are localized to the business timezone (configurable; default Asia/Bangkok) before hour extraction. Transactions between the configured peak window (default 17:00–21:00) are flagged `Peak`; all others `Off-Peak`. Off-peak promotions target demand stimulation during low-traffic periods, preserving full-margin sales during peak hours.

Inventory estimation: In the absence of live inventory feeds — standard in SME deployments — remaining stock is estimated from a velocity-based replenishment model:

$$\text{initial_stock_est} = \frac{\text{total_quantity_sold}}{T} \times R \times \text{buffer_multiplier}$$

$$\text{remaining_stock} = \text{initial_stock_est} - \text{recent_demand}_R$$

where T is the observation window in days, R is the assumed replenishment cycle (default 14 days), and `recent_demand_R` is quantity sold in the last R days. This formulation correctly identifies fast-moving products with accelerating recent demand — rather than low-volume products — as candidates for stock-out exclusion. Products with remaining_stock ≤ 5 units are flagged out-of-stock and excluded from bundle recommendations.

Campaign matching applies a rule engine combining cluster tier, category affinity, and context flags to produce an initial campaign assignment, subsequently refined by Layer 4's uplift and margin analysis.

4.4 Layer 4 — Uplift Modeling and CFO Guardrail

Input: Enriched customer records from Layers 1–3 **Output:** `uplift_score`, `recommended_action`, `expected_profit` per customer; champion/challenger model comparison report

4.4.1 Uplift Model (T-Learner)

The pipeline implements a T-learner (two-model approach) [11]: two LightGBM classifiers are trained independently on disjoint subgroups:

- **Treatment model:** trained on customers who received at least one campaign (`was_treated = 1`)
- **Control model:** trained on customers who received no campaign (`was_treated = 0`)

Per-customer uplift is estimated as:

$$\text{uplift}_i = \hat{P}(\text{convert} \mid \text{treatment})_i - \hat{P}(\text{convert} \mid \text{control})_i$$

Positive uplift indicates incremental expected purchase attributable to the campaign. Negative uplift indicates the customer would likely purchase regardless, making campaign spend wasteful.

Important caveat: T-learner uplift is only causally interpretable when treatment assignment is randomized. In the absence of a randomized experiment — typical at first SME deployment — the model produces a *response-propensity heuristic*, not a causal estimate. The 5% ongoing holdout control group enables properly randomized uplift measurement to accumulate over time, converting the system from a batch propensity scorer into a growing randomized experiment.

Label sourcing — real vs. proxy: The pipeline supports two label modes selected automatically at runtime:

1. **Real promotion response mode** (preferred): When `is_treated` and `post_promo_visit` columns are present in the POS file, the system uses these as treatment and conversion labels. When a `marketing_campaign.csv` file is additionally present with `AcceptedCmp1` – `AcceptedCmp5` and `Response` columns, three additional behavioral features are activated: `income`, `total_spend`, and `times_accepted`. Note that `AcceptedCmp` columns record whether a customer *accepted* a campaign offer — they are observational outcomes, not randomized treatment assignments. Uplift estimates from this source reflect self-selection bias and should be interpreted as directional signals rather than causal effects.
2. **RFM proxy mode** (fallback): When no promotion response file is available — the common case for first-time SME deployments — treatment labels are derived from RFM decile rank (customers above the median decile are proxied as treated) and conversion labels from repeat purchase frequency. This mode introduces circularity: frequency contributes to both the treatment proxy and the outcome proxy. It should be interpreted strictly as a propensity-to-respond ranking, not as an uplift estimate. Upgrading to real label mode as promotion response data accumulates is the intended operational path.

The feature sets differ between modes:

Feature	Proxy Mode	Real Label Mode
`total_price` (AOV)	✅	✅
`quantity`	✅	✅
`is_campaign_spike`	✅	✅
`rfm_decile_segment`	✅	✅
`frequency`	✅	✅
`income`	❌	✅
`total_spend`	❌	✅
`times_accepted`	❌	✅

A 5% holdout control group is preserved in every deployment cycle for ongoing A/B measurement of realized incremental lift. As real promotion response data accumulates through partner deployments, operators automatically transition from proxy mode to real label mode without pipeline reconfiguration.

4.4.2 Decision Rules

Final action assignment integrates uplift score, RFM decile, AOV, and context flags through a priority-ordered rule set:

Priority	Condition	Assigned Action
1	uplift < −0.04 OR AOV < 100	Do Nothing
2	CFO Guardrail fails (net_margin < floor OR Δprofit ≤ 0)	Do Nothing (CFO Guardrail)
3	uplift > 0.08 AND bundle available	Push Promotion (Bundle)
4	RFM decile ≥ 9	VIP Early Access
5	RFM decile ≥ 8 AND P(control) ≥ 0.4	VIP Reward: X2 Points
6	Off-Peak AND 100 ≤ AOV ≤ 800 AND RFM ≥ 4	Off-Peak Special
7	Default	Standard Retention

4.4.3 CFO Guardrail

Before finalizing any campaign assignment, two per-customer financial checks are applied:

1. Net margin floor check (per campaign type):

$$\text{gross_revenue}_i = \text{AOV}_i \times (1 - d)$$

$$\text{net_profit}_i = \text{gross_revenue}_i - \text{AOV}_i \times c_{\text{COGS}} - \text{AOV}_i \times r_{\text{promo}}$$

$$\text{net_margin}_i = \frac{\text{net_profit}_i}{\text{gross_revenue}_i}$$

where d is the campaign discount rate and r_{promo} is the promo cost rate (both specific to the assigned action type). If `net_margin_i < min_allowable_net_margin` (default 0.15), the action is overridden to `Do Nothing`.

2. Per-customer incremental profit check:

$$\Delta\text{profit}_i = [\text{gross_margin}_i \times \hat{P}(\text{convert} \mid \text{treated})_i - \text{promo_cost}_i] - [\text{gross_margin}_i \times \hat{P}(\text{convert} \mid \text{control})_i]$$

If $\Delta\text{profit}_i \leq 0$, the campaign is expected to cost more than the incremental revenue it generates for customer i, and the action is overridden to `Do Nothing`. This check is genuinely per-customer because Δprofit_i depends on each customer's predicted uplift score and individual AOV.

Both checks must pass for a campaign action to be dispatched. This dual-gate design ensures both structural profitability (margin floor) and incremental profitability (positive expected return above baseline) per customer at inference time.

Prior work on profit-driven uplift modeling [27, 28] has established the value of integrating financial objectives into campaign targeting. The CFO Guardrail's contribution is implementing this as a hard, config-driven constraint enforced at inference time — eliminating the manual organizational review step that exists as a latency bottleneck in conventional SME deployments.

All financial parameters (`cogs_rate`, `min_allowable_net_margin`, `promo_cost_rates`) are configurable per business in `business_config.py`, enabling deployment across operators with different cost structures without code changes.

4.4.4 Champion-Challenger Protocol

Model versioning follows a Champion-Challenger pattern. A challenger model (trained on the latest data) is automatically promoted to Champion production status if either condition holds:

- Challenger expected profit exceeds Champion by ≥ 3% (growth gate), OR
- Recession mode is active — defined as current mean spend < 65% of baseline — AND challenger `Do Nothing` rate exceeds 50%, indicating capital preservation is appropriate

Both champion and challenger models are serialized with timestamps and retained in an archive for full reproducibility and rollback capability.

4.5 Layer 5 — LLM Executive Report

Input: Final campaign assignments and financial projections **Output:** Five-section plain-language executive summary

We deploy `llama3.2:3b` locally via Ollama (temperature = 0.0 for deterministic output; max 3,500 tokens). Input data is pre-formatted as structured text before the LLM call to reduce parsing ambiguity and improve output consistency. The generated report covers:

(1) segment overview, (2) campaign recommendations, (3) financial projections, (4) execution plan, and (5) post-campaign forecast.

A post-generation validation step cross-checks segment names, customer counts, and profit figures in the LLM output against the source payload. Discrepancies trigger an appended warning block rather than silent failure — preserving transparency while avoiding pipeline interruption.

Local inference ensures no customer transaction data is transmitted to external services, addressing data privacy requirements relevant to SME operators subject to regional data protection regulations.

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5. Experiments

5.1 Datasets

Two datasets were used to evaluate the framework across complementary dimensions: an F&B-calibrated synthetic dataset for methodology validation, and a real-world dataset with ground-truth promotion response labels for uplift model validation.

Dataset 1 — Synthetic Restaurant POS Dataset

A synthetic dataset generated to simulate a representative Thai F&B SME environment, calibrated to industry-typical purchase patterns including peak-hour dining, repeat weekly visits, and menu-level product affinity. The dataset contains 9,272 *transaction line items* (each line item is a single menu item ordered in a visit; a single customer visit may produce multiple line items) from 200 unique customers across a 6-month observation window (May–November 2026). This distinction is important: RFM Frequency and AOV are computed at the line-item level before aggregation to the customer level, which produces higher per-line-item AOV and frequency counts than a bill-level aggregation would. Menu items span five categories: Main Course, Dessert, Beverage, Appetizer, and Set Menu. Customer behavior is stratified across four generator segments — Champions (30), Regular (60), Occasional (70), and At-Risk (40) — with visit frequencies and spend distributions calibrated per segment. A 30% treatment group (60 customers) was randomly assigned via the generator to receive LINE-delivered promotions with measured response (`is_treated`, `post_promo_visit`), providing randomized treatment labels for Layer 4 real label mode. The treatment assignment is randomized at the

generator level, making this dataset suitable for T-learner uplift evaluation on a simulated randomized experiment. This dataset directly represents the target SME deployment context.

Dataset 2 — Marketing Campaign Dataset

A public dataset from Kaggle [26] containing customer-level marketing data for 2,240 customers, including cumulative spending across product categories (`MntWines`, `MntFruits`, etc.), campaign acceptance history (`AcceptedCmp1` – `AcceptedCmp5`), and a final response indicator (`Response`). Because the original dataset does not include transaction-level records or timestamps, a synthetic transaction representation was constructed for pipeline input: each customer's total category spend was disaggregated into individual line items (one per spending category with spend > 0), producing 4,899 transaction records from 905 active customers (customers with at least one recorded purchase). Timestamps for Layer 3 context detection are synthetically generated and carry no real-world temporal meaning for this dataset; Layer 3 context results from this dataset are informational only. After preprocessing, this dataset activates Layer 4's real label mode using `AcceptedCmp1` – `AcceptedCmp5` and `Response` as observational treatment/conversion proxies. All values are in USD (not THB). Note that `AcceptedCmp` columns record voluntary acceptance of offers — they are observational outcomes subject to self-selection, not randomized treatment assignments. Uplift estimates from this dataset reflect directionality rather than causal effect.

5.2 Stress Test Scenarios

To assess robustness under distribution shifts common in SME operations, the pipeline was additionally evaluated against six synthetic stress scenarios generated from the primary dataset:

Scenario	Description	Key Challenge	Pass Criterion
`mega_sale`	Flash sale / Black Friday spike	Extreme volume, inflated AOV	Campaign spike flag rate ≥ 80% of spike transactions
`recession`	Economic downturn simulation	Suppressed spend, baseline drift	Recession mode activates; Do Nothing rate ≥ 50%
`bulk_shift`	Bulk buyer behavior change	Heavy-tail outlier distribution shift	IQR filter removes outliers; segment centroids shift < 20%
`cold_start`	New market or new location	Sparse transaction history (<50 customers)	k = 2 fallback; pipeline produces ≥ 1 campaign action
`off_peak`	Off-peak hour concentration	Time context edge case	Off-peak context detected; off-peak action rate > 60%
`data_noise`	Corrupted timestamps, missing fields	Data quality and robustness	Zero unhandled exceptions; < 5% rows silently dropped

5.3 Evaluation Metrics

Metric	Layer	Description
Silhouette score	L1	Cluster separation quality (−1 to +1)
Optimal k	L1	Dynamic cluster count selection
Association rule lift	L2	Bundle recommendation strength
Uplift score distribution	L4	Treatment effect signal quality
Expected net profit per customer	L4	Per-head financial value of recommendation
Champion-Challenger profit delta (%)	L4	Model update quality
Margin compliance rate (%)	L4	CFO guardrail filter effectiveness
Stress test pass rate	All	Robustness under distribution shift

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6. Results

6.1 Customer Segmentation — Synthetic Restaurant Dataset

The dynamic cluster selection procedure identified **k = 2** as optimal for the restaurant dataset (silhouette score = **0.679**), producing two customer tiers with strong separation reflecting the typical F&B SME customer structure: a high-engagement Champions group and a broader At-Risk group.

Segment	Customers	% of Total	AOV*	Recency	Frequency	Net Profit/Head	Campaign
Champions	30	15%	฿3,094	4.8 days	166.4	฿1,522	VIP Bundle
At-Risk	170	85%	฿474	19.5 days	25.2	฿166	Bundle −15%
Total	**200**						

AOV = average total_price per transaction line item. Customer-level weighted mean AOV across all 200 customers is ฿179.

Total projected campaign profit: ฿74,439 (gross revenue uplift before promo cost deduction; net incremental profit after guardrail filtering: ฿2,162, see Section 6.7) | Revenue lift vs baseline: +3.29%

The Champions segment, representing 15% of customers, generates a net profit per head of ฿1,522 — 9.2× higher than the At-Risk segment — despite being a small minority. This ratio is consistent with Pareto dynamics commonly observed in F&B loyalty programs, where repeat high-frequency customers disproportionately drive profitability.

The silhouette score of **0.679** is the highest observed across all evaluations, reflecting that restaurant POS data — characterized by regular repeat visits and stable spend patterns — produces particularly well-separated RFM clusters. The IQR outlier filter had minimal impact (< 0.5% records removed), consistent with the controlled synthetic generation process.

6.2 Customer Segmentation — Marketing Campaign Dataset

For the Marketing Campaign dataset (905 customers — larger customer count than Dataset 1 though with fewer line items per customer), dynamic selection identified **k = 3** as optimal (silhouette score = **0.347**).

Segment	Customers	% of Total	AOV	Net Profit/Head	Campaign
Champions (High)	323	35.7%	$1,323	$635	VIP Early Access
Champions (Low)	456	50.4%	$230	$111	VIP Reward: X2 Points
Promising	126	13.9%	$229	$80	Bundle −15%
Total	**905**				

Total projected campaign profit: $265,650 USD

The lower silhouette score (0.347 vs. 0.679) reflects the aggregate-level nature of the marketing campaign data, which produces less geometrically distinct RFM clusters than transaction-level POS data. The pipeline correctly adapts to k = 3, demonstrating that dynamic cluster selection prevents artificial fragmentation. Champions (High) AOV of $1,323 is the highest single-segment AOV observed across both datasets. This dataset activates Layer 4's real label mode using `AcceptedCmp` and `Response` columns; however, because campaign acceptance is self-selected (not randomized), uplift estimates from this mode are observational signals rather than causal effects (see Section 4.4.1 caveat).

6.3 Cross-Dataset Comparison

Metric	Restaurant POS (F&B)	Marketing Campaign
Line items	9,344	4,899 (synthesised from aggregates)
Customers	200	905
Observation window	6 months (real timestamps)	Aggregate (synthetic timestamps)
Optimal k	2	3
Silhouette score	**0.679**	0.347
AUUC (T-learner)	**0.2351**	N/A (observational labels)
Top segment AOV	฿3,094	$1,323
Top segment Net/Head	฿1,522	$635
Total projected profit	**฿74,439**	**$265,650 USD**
Revenue lift	**+3.29%**	N/A
Label mode	Real (is_treated, randomised)	Observational (AcceptedCmp)

The restaurant dataset produces higher silhouette (0.679 vs 0.347), consistent with the expectation that repeat-visit F&B data with randomized treatment assignment generates tighter RFM clusters and interpretable uplift signals. The marketing campaign dataset provides a larger, real-world customer base for segmentation validation with observational promotion response data. Together, the two datasets validate pipeline behavior across transaction-level F&B data (Dataset 1) and aggregate retail data (Dataset 2).

6.4 Champion-Challenger Performance

The Champion-Challenger evaluation compares a freshly trained Challenger model against the incumbent Champion on the same customer population. On first deployment (no saved Champion), the Challenger is accepted as the initial production model.

In subsequent cycles, the Challenger is promoted if expected profit improvement ≥ 3%. In recession mode (current mean spend < 65% of baseline), promotion is triggered if the Challenger's `Do Nothing` rate exceeds 50%, prioritizing capital preservation over growth. The protocol ensures model updates are financially motivated rather than statistically arbitrary.

On the 200-customer synthetic dataset, LightGBM with n=30 estimators converges deterministically — different random seeds produce identical model outputs — resulting in zero measured improvement variance across seeds (see Section 6.7). This is an honest limitation of the small synthetic dataset, not an indication of model superiority; it confirms the need for real pilot data to demonstrate non-trivial Champion-Challenger dynamics.

6.5 CFO Guardrail Behavior

The dual-gate guardrail (margin floor check + per-customer incremental profit check) operates as the final financial filter before campaign dispatch. On the synthetic restaurant dataset:

- **142 of 200 customers (71.0%)** are overridden to `Do Nothing` by the guardrail
- Guardrail-blocked customers have near-zero predicted incremental profit (Δprofit ≤ 0), confirming the filter is functioning correctly rather than over-blocking
- **Ablation (Section 6.7):** removing the guardrail produces total expected profit of **–฿533** vs the full pipeline's **฿2,162** — a ฿2,695 deterioration — demonstrating that the guardrail is the primary mechanism preventing money-losing campaigns on this synthetic dataset

The block rate (71.0%) on this synthetic dataset reflects the small absolute AOV (customer-level average: ฿179; segment line-item AOVs range from ฿474 to ฿3,094 as shown in Table 6.1) and the modest control-group conversion signal, which makes the per-customer incremental profit threshold moderately stringent. In real deployments

with larger populations and accumulated control-group response data, the guardrail block rate is expected to be lower while remaining the primary financial safety mechanism.

6.6 Stress Test Robustness

Scenario	Pipeline Status	Measured Outcome	Criterion Met?
`mega_sale`	✅ Pass	Spike flag rate = 100% of spike transactions	≥ 80% ✓
`recession`	✅ Pass	Recession mode activates; Do Nothing rate = 87%	≥ 50% ✓
`bulk_shift`	✅ Pass	IQR removes outliers; centroid shift < 8%	< 20% ✓
`cold_start`	✅ Pass	k = 2 fallback; 1+ campaign actions produced	≥ 1 ✓
`off_peak`	✅ Pass	Off-peak context flag rate = 72%	> 60% ✓
`data_noise`	✅ Pass	0 unhandled exceptions; 2.1% rows dropped	0 exc, < 5% ✓

Stress test pass rate: 6/6 (100%) — all quantitative criteria met

"Pass" is defined here as meeting each scenario's pre-specified quantitative criterion, not merely "pipeline did not crash." Criteria were defined prior to running stress tests (see Section 5.2).

6.7 Comparative Evaluation — Baselines, Ablation, and Qini

6.7.1 Uplift Model Quality: Qini Curve and AUUC

The T-learner uplift model is evaluated using the Qini curve and Area Under the Uplift Curve (AUUC) metric, computed on the synthetic dataset where treatment assignment is randomized (60 treated, 140 control; 160 total converters including control-group spontaneous returns).

AUUC = 0.2351 against a random-targeting baseline, confirming that the model ranks customers better than random at identifying responders. The Qini curve reaches 0.4233 at $\tau = 0.30$, meaning that by targeting only the top 30% of customers (ranked by predicted uplift), the pipeline captures 42% of total incremental conversion — a meaningful efficiency gain over random targeting at the same coverage.

Important interpretation note: With the updated data generator, control customers now have a baseline return rate (spontaneous visits without promotion), enabling the T-learner to distinguish genuine uplift from baseline propensity. The AUUC of 0.2351 reflects the model's ability to rank customers by incremental response, which is operationally useful for targeting decisions. Genuine causal uplift estimation still requires accumulated holdout data from the 5% ongoing control group for rigorous validation.

6.7.2 Baseline Comparison

Net incremental profit (after promo costs, post guardrail)

Method	Total Expected Profit	Δ vs Full Pipeline
Full Pipeline (uplift + guardrail + context)	**฿2,162**	—
Broadcast All Customers (standard campaign)	฿2,109	+฿53 (+0.3%)
RFM-Only (no uplift model)	−฿1,334	+฿3,496
Random 30% Targeting	฿744	+฿1,418

Note: The 'RFM-Only (no uplift model)' baseline in Table 6.7.2 uses prob_control as conversion estimate (which is near-zero for purely promo-driven customers, making promo costs dominate). The '− Guardrail − Context' row in Table 6.7.3 differs because it uses uplift scores but removes both guardrail and context — an architectural combination, not a strict RFM baseline.

The full pipeline marginally outperforms broadcast (+0.3%), confirming that on a small 200-customer synthetic dataset the uplift model's contribution over naive broadcast is modest. The more significant comparison is against RFM-only and random baselines, which both produce substantially lower — or negative — expected profits. The RFM-only result (−฿1,334) demonstrates that assigning campaigns based on segment alone,

without an uplift model to calibrate expected return, frequently dispatches money-losing campaigns. At larger scale (thousands of customers and accumulated real uplift signal), the differential between full pipeline and broadcast is expected to widen.

6.7.3 Ablation Study

Note: The 'No Uplift Model' row reflects a structural dependency: without uplift scores, the per-customer incremental profit $\Delta profit_i = 0 - promo_cost < 0$ for all customers, causing the CFO Guardrail to block all campaigns. The Δ฿2,162 therefore represents the combined contribution of the uplift model enabling the guardrail to operate selectively, not the uplift model's independent contribution.

Configuration Removed	Expected Profit	Δ vs Full	Interpretation
Full Pipeline	฿2,162	—	Baseline
− CFO Guardrail	−฿533	**−฿2,695**	Guardrail prevents all money-losing campaigns
− Context Layer (Layer 3)	฿2,162	±฿0	No material effect on synthetic dataset (all items in stock)
− Uplift Model (Layer 4)	฿0	−฿2,162	Without uplift, incremental profit = −promo_cost for all → guardrail blocks all
− Guardrail + Context	฿0	−฿2,162	Combined RFM-only: same result as no-uplift (guardrail still fires)

Two findings stand out:

1. **The CFO Guardrail is the primary mechanism preventing money-losing campaigns on this synthetic dataset**, preventing ฿2,695 in negative expected profit. This strongly validates the architectural decision to embed financial constraint checking inside the ML decision loop.

2. **Context detection has no material effect on this synthetic dataset**, as the velocity-based stock estimator flags no items as low-stock under the controlled generation process. In real deployments with actual stock-outs and temporal demand variation, Layer 3 context effects are expected to be more significant.

6.7.4 Multi-Seed Stability

The pipeline was run with 10 different LightGBM random seeds (0–9) to assess model stability:

Metric	Mean	SD	Stability
Total expected profit	฿2,162	±฿0	Fully deterministic
AUUC	0.1487	±0.0000	Fully deterministic
Guardrail blocked	142.0	±0.0	Fully deterministic

Zero variance across seeds indicates deterministic convergence: LightGBM with n_estimators=30 on a 200-customer dataset finds the same splits regardless of random seed. This is an honest limitation of evaluating on a small synthetic dataset — it does not mean the model has zero variance in general, but rather that the dataset is too small to exhibit training-set sampling variance across seeds. This confirms the need for real pilot data to conduct statistically meaningful multi-seed evaluation. For the purpose of this paper, deterministic convergence ensures full reproducibility of all reported results.

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7. Discussion

7.1 The CFO Guardrail as Organizational Boundary Automation

The most significant novel contribution of this framework is **architectural**: the placement of a financial constraint validator inside the ML decision loop rather than as a downstream organizational review step.

Prior work on profit-driven uplift modeling [27, 28] has established the value of integrating financial objectives into campaign targeting. Gubela et al. [27] and Devriendt et al. [28] show that incorporating revenue and cost structure into the uplift optimization objective improves financial outcomes over pure response-probability targeting. Our

CFO Guardrail extends this direction by implementing financial validation as a *hard constraint* — rather than a soft objective — enforced per customer at inference time, and exposing all cost parameters through a single operator-editable configuration file.

In conventional enterprise deployments, data science teams generate recommendations reviewed by finance or marketing operations before deployment. This creates latency (typically days to weeks in SME contexts where reviews are informal), organizational friction, and the risk that financially sound recommendations are delayed or discarded for non-quantitative reasons.

By parameterizing financial constraints — COGS rate, margin floor, discount rates — and applying them per customer at inference time, the system eliminates the review step for routine campaigns. Edge cases surface naturally through the `Do Nothing` classification rather than requiring human escalation. The `business_config.py` file serves simultaneously as the operator's financial policy document and the system's decision boundary, creating a direct and auditable link between business rules and campaign outputs.

The ablation result (Section 6.7.3) quantifies this contribution empirically: the CFO Guardrail prevents ฿2,695 in negative expected profit on the 200-customer synthetic dataset, confirming that it is the primary mechanism preventing money-losing campaigns on this synthetic dataset.

7.2 Dynamic Cluster Selection as SME Adaptation

The results across both datasets illustrate why hardcoding cluster count is inappropriate for SME deployment contexts. The restaurant dataset (200 customers, dense repeat-visit behavior) produces optimal k = 2 with silhouette = 0.679; the Marketing Campaign dataset (905 customers, aggregate-level data) produces k = 3 with silhouette = 0.347. A system that hardcoded k = 3 for the restaurant dataset would artificially split a naturally binary customer structure, while hardcoding k = 2 for the marketing dataset would collapse meaningful behavioral distinctions.

The dynamic selection procedure, selecting k by maximizing silhouette score over $k \in \{2, \ldots, 5\}$, automatically scales granularity to data density. This is a key enabler of the minimal-input design principle: operators do not need to tune segmentation parameters.

7.3 Accessibility versus Accuracy Trade-offs

The framework makes deliberate simplifications to maintain accessibility. These trade-offs are transparent:

Inventory estimation from sales velocity introduces estimation error in the absence of live stock feeds — typical for SME POS systems. Operators with inventory management systems can replace the velocity-based estimator with direct stock queries through the `stock_restock_cycle_days` and `stock_buffer_multiplier` configuration parameters.

Treatment labels derived from the generator's randomized assignment in Dataset 1, and from observational campaign acceptance records in Dataset 2 — both requiring caution in causal interpretation as discussed in Sections 4.4.1 and 5.1 — is a necessary concession for SME settings where historical A/B data is unavailable. The 5% ongoing control group enables proper causal measurement going forward, converting the system from a batch estimator into an accumulating randomized experiment over time.

Local LLM inference (3B parameter model) produces adequate executive summaries at zero ongoing cost and with full data privacy, but may produce less nuanced analysis than larger hosted models. The structured output validation layer partially compensates by catching factual errors in segment names, customer counts, and profit figures.

7.4 Extensibility

The pipeline architecture is designed for vertical expansion. The six-column input schema is generic enough to accommodate any transactional business. The `business_config.py` central configuration enables parameter tuning without code changes. Planned extensions include:

- **Event-driven campaign prediction:** Incorporating scheduled events (festivals, public holidays, promotional calendars) as Layer 3 context signals for pre-emptive campaign planning
- **POS-CRM-LINE integration:** Connecting the pipeline output directly to LINE-based ordering and delivery systems, enabling automated campaign dispatch through existing customer communication channels
- **Multi-location aggregation:** Pooling POS streams from multiple outlets with site-level context flags for chain operators
- **Service business adaptation:** Mapping `product_name` to service SKUs for beauty, automotive, and professional services verticals

7.5 Limitations and Future Work

Evaluation on synthetic data: Dataset 1 is synthetically generated from known segment parameters. The silhouette score (0.679) and k=2 selection reflect the generator's structure rather than discovering latent behavioral segments in real transaction data. Deterministic multi-seed convergence confirms the 200-customer dataset is too small for variance quantification. Evaluation on real pilot deployment data is the critical next step.

T-learner operates on synthetic labels: Control customers in Dataset 1 now have a baseline return rate (spontaneous visits without promotion), enabling the T-learner to produce non-trivial prob_control estimates. However, AUUC = 0.2351 still reflects model behavior on synthetic labels from a known generator. Genuine causal uplift estimation requires accumulated holdout response data from the 5% control group preserved in each deployment cycle.

Observational labels in Dataset 2: Marketing campaign acceptance (`AcceptedCmp`) reflects customer self-selection, not randomized assignment. Uplift estimates from Dataset 2 are observational and subject to confounding. They should not be interpreted as causal effects.

Baselines gap on small dataset: The full pipeline's margin over broadcast (+0.3%) is small at n=200, which is expected given the small dataset. The material differential appears in the comparison to RFM-only (−฿1,334 vs ฿2,162), confirming that the guardrail and uplift model together prevent money-losing campaigns.

Context layer limitation: The velocity-based stock estimator showed no effect on the synthetic dataset (no items flagged low stock). Layer 3's contribution is expected to be more material in real deployments with genuine stock variability and temporal demand patterns.

Association rule lift: Lift values in the synthetic dataset reflect structured menu co-purchase patterns by construction. Real-world deployment is expected to produce lower but more actionable lift values as order diversity increases.

Future work will examine: (1) real pilot deployment data from partner restaurant operators with 5% control groups; (2) online learning to update RFM and uplift models continuously; (3) multi-armed bandit formulations for campaign action selection; (4) Qini-curve-based uplift objective integration to replace the current net-margin floor with a continuous financial objective in the LightGBM training loss.

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8. Conclusion

This paper presented the ML Campaign Engine, a five-layer MLOps pipeline that enables small and medium enterprises in retail and food service to generate financially validated, per-customer campaign recommendations from standard POS transaction exports. The system requires no dedicated data science expertise, no additional data infrastructure beyond an existing POS system, and produces outputs in plain-language executive summary format suitable for direct operator consumption.

The dual-gate CFO Guardrail — enforcing a net margin floor and a per-customer incremental profit check at inference time — extends prior work on profit-driven uplift modeling [27, 28] by implementing financial validation as a hard, config-driven constraint embedded in the ML decision loop, eliminating the manual organizational review step typical of conventional deployments. Ablation analysis demonstrates that the CFO Guardrail is the primary mechanism preventing money-losing campaigns on this synthetic dataset, preventing ฿2,695 in negative expected profit.

Evaluation on two datasets confirms context-appropriate behavior: the synthetic restaurant dataset (200 customers) yields k = 2 with silhouette = 0.679, AUUC = 0.2351, gross revenue uplift of ฿74,439 and net incremental profit after financial filtering of ฿2,162; the public marketing campaign dataset (905 customers) yields k = 3 with silhouette = 0.347 and $265,650 USD in projected profit. Dynamic cluster selection correctly adapts to both data structures. All six stress-test scenarios pass quantitative pass criteria. The T-learner uplift model outperforms random targeting (AUUC = 0.2351), though causal interpretation requires accumulated holdout data from the 5% ongoing control group.

Honest limitations include the small synthetic dataset, deterministic model convergence at n=200, and the absence of real pilot data. The next step is real deployment with partner restaurant operators and a proper randomized holdout to establish realized lift beyond model-estimated projections.

The framework establishes a reproducible, extensible baseline for democratizing campaign intelligence for SMEs, and provides a validated technical foundation for partner deployment and real-world A/B evaluation.

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Appendix A — Pipeline Configuration Parameters

All business parameters are centralized in `src/business_config.py`. Operators adjust this file to configure the pipeline for their cost structure without modifying pipeline code.

Parameter	Default	Description
`cogs_rate`	0.50	Cost of goods as fraction of revenue
`min_allowable_net_margin`	0.15	CFO floor — campaigns below this margin → Do Nothing
`control_group_rate`	0.05	A/B holdout fraction reserved for causal measurement
`gate_growth_target_pct`	3.0	Challenger promotion threshold (%)
`recession_threshold`	0.65	Spend ratio below which recession mode activates
`peak_hour_start`	17	Start of peak window (24h, inclusive)
`peak_hour_end`	21	End of peak window (24h, inclusive)
`business_tz`	Asia/Bangkok	Business timezone for peak detection
`low_basket_threshold`	100	AOV below this → Do Nothing
`high_uplift_threshold`	0.08	Uplift above this → bundle promotion
`vip_rfm_threshold`	8	RFM decile ≥ this → VIP reward
`stock_buffer_multiplier`	1.5	Estimated stock covers this multiple of avg demand per restock cycle
`stock_restock_cycle_days`	14	Assumed replenishment cycle in days
`low_stock_threshold`	5	Remaining stock ≤ this → exclude from bundle

Appendix B — Stress Test Scenario Definitions

Scenario	Volume	AOV Distribution	Temporal Pattern	Primary Test
`mega_sale`	3× normal	Right-skewed, high AOV	Peak-concentrated	Spike detection, guardrail under inflated margins
`recession`	0.6× normal	Left-shifted, low AOV	Uniform	Recession mode activation, Do Nothing rate
`bulk_shift`	Normal	Extreme right tail	Uniform	IQR outlier isolation
`cold_start`	0.2× normal	Normal	Uniform, short window	k = 2 fallback, sparse data handling
`off_peak`	Normal	Normal	00:00–16:00 concentrated	Off-peak context detection
`data_noise`	Normal	Normal + NaN injection	10% corrupted timestamps	Error handling, graceful degradation

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